

Independence study of MATH 655; Week 3

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How to get D'Alembert's formula

1d wave equation is defined by following

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0, & x \in \mathbb{R}, t > 0, \\ u(x, 0) = g(x), \\ u_t(x, 0) = h(x). \end{cases} \quad (*)$$

First factor the PDE to

$$(\partial_t - c\partial_x)(\partial_t + c\partial_x)u = 0.$$

Since

$$\begin{aligned} (\partial_t - c\partial_x)(\partial_t + c\partial_x)u &= 0 \\ &= u_{tt} + cu_{tx} - cu_{xt} - c^2 u_{xx} \end{aligned}$$

By Calc 3 we know $u_{tx} = u_{xt}$, if u is smooth.

$$= u_{tt} - c^2 u_{xx} = 0.$$

Now let

$$\delta = x + ct, \quad \xi = x - ct,$$

where δ means wave travel to the left,

ξ means wave travel to the right.

Since 1 dimensional wave can only travel for this two direction.

Rewrite

$$u(x, t) = v(\delta, \xi),$$

i.e.

$$u(x, t) = v(\delta, \xi) = v(x + ct, x - ct).$$

WTK: u_x

By chain rule:

$$u_x = v_\delta \delta_x + v_\xi \xi_x.$$

Since

$$\delta_x, \xi_x = 1.$$

So

$$\begin{aligned} u_x &= v_\delta + v_\xi. \\ u_{xx} &= v_{\delta\delta} + 2v_{\delta\xi} + v_{\xi\xi}. \end{aligned} \tag{1}$$

WTK: u_t

$$u_t = v_\delta \delta_t + v_\xi \xi_t.$$

Since

$$\begin{aligned} \delta_t &= c, & \xi_t &= -c. \\ u_t &= cv_\delta - cv_\xi. \\ u_{tt} &= c^2 v_{\delta\delta} - 2c^2 v_{\delta\xi} + c^2 v_{\xi\xi}. \end{aligned} \tag{2}$$

Bring (1), (2) into (*)

$$u_{tt} - c^2 u_{xx} = (c^2 v_{\delta\delta} - 2c^2 v_{\delta\xi} + c^2 v_{\xi\xi}) - (c^2 v_{\delta\delta} + 2c^2 v_{\delta\xi} + c^2 v_{\xi\xi}).$$

Simplified:

$$u_{tt} - c^2 u_{xx} = -4c^2 v_{\delta\xi}.$$

i.e.

$$-4c^2 v_{\delta\xi} = 0 \implies \boxed{v_{\delta\xi} = 0}.$$

Solve

$$v_{\delta\xi} = 0.$$

$$(v_\delta)_\xi = 0$$

also tells you function v_δ only depends on δ .

So let

$$v_\delta = A'(\delta).$$

Integrate to δ , we obtain:

$$v(\delta, \xi) = A(\delta) + B(\xi).$$

Hence

$$\boxed{u(x, t) = v(x + ct, x - ct) = A(x + ct) + B(x - ct).} \tag{3}$$

Now WTK: A, B .

Use

$$u(x, 0) = g(x).$$

Let $t = 0$ for (3), we have

$$u(x, 0) = A(x) + B(x) = g(x). \quad (4)$$

Use

$$u_t(x, 0) = h(x).$$

Note that

$$u_t(x, t) = cA'(x + ct) - cB'(x - ct) = h(x).$$

Let $t = 0$,

$$u_t(x, 0) = cA'(x) - cB'(x) = h(x). \quad (5)$$

Combine (4) and (5) we have:

$$\begin{cases} A(x) + B(x) = g(x), \\ A'(x) - B'(x) = \frac{h(x)}{c}. \end{cases}$$

Take $\frac{d}{dx}$ of (4) make it easier,

$$\begin{cases} A'(x) + B'(x) = g'(x), & \text{(a)} \\ A'(x) - B'(x) = \frac{h(x)}{c}. & \text{(b)} \end{cases}$$

(a) + (b) we have

$$2A'(x) = g'(x) + \frac{h(x)}{c}.$$

$$A'(x) = \frac{1}{2}g'(x) + \frac{1}{2c}h(x).$$

Plug $A'(x)$ to (a) we have

$$B'(x) = g'(x) - \frac{1}{2}g'(x) - \frac{1}{2c}h(x)$$

$$= \frac{1}{2}g'(x) - \frac{1}{2c}h(x).$$

Thus,

$$\begin{cases} A'(x) = \frac{1}{2}g'(x) + \frac{1}{2c}h(x), \\ B'(x) = \frac{1}{2}g'(x) - \frac{1}{2c}h(x). \end{cases}$$

Integrate $A'(x)$ with dx .

$$A(x) = \frac{1}{2}g(x) + \frac{1}{2c} \int_a^x h(y) dy + C_1.$$

Integrate $B'(x)$ with dx .

$$B(x) = \frac{1}{2}g(x) - \frac{1}{2c} \int_a^x h(y) dy + C_2.$$

Back to (3) we conclude

$$u(x, t) = \frac{1}{2}g(x + ct) + \frac{1}{2c} \int_a^{x+ct} h(y) dy + \frac{1}{2}g(x - ct) - \frac{1}{2c} \int_a^{x-ct} h(y) dy.$$

Recall calc 1 we have

$$\int_a^{x+ct} h(y) dy - \int_a^{x-ct} h(y) dy = \int_{x-ct}^{x+ct} h(y) dy.$$

Therefore, D'Alembert Formula:

$$u(x, t) = \frac{1}{2} [g(x + ct) + g(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} h(y) dy.$$

Circular Means and the Euler–Poisson–Darboux Equation in $n = 2$

We consider the two-dimensional wave equation

$$u_{tt} - \Delta u = 0, \quad x \in \mathbb{R}^2, \quad t > 0.$$

The goal is to show that the circular mean of u satisfies the Euler–Poisson–Darboux equation.

1. Where does the circular mean come from?

For a fixed point $x \in \mathbb{R}^2$ and radius $r > 0$, the circle centered at x with radius r is

$$\partial B(x, r) = \{y \in \mathbb{R}^2 : |y - x| = r\}.$$

The average of u on this circle is defined by

$$U(x; r, t) := \frac{1}{|\partial B(x, r)|} \int_{\partial B(x, r)} u(y, t) dS_y.$$

In two dimensions,

$$|\partial B(x, r)| = 2\pi r.$$

We parameterize the circle by

$$y = x + r(\cos \theta, \sin \theta), \quad 0 \leq \theta \leq 2\pi.$$

Since the arclength element is

$$dS_y = r d\theta,$$

we get

$$U(x; r, t) = \frac{1}{2\pi r} \int_0^{2\pi} u(x + r(\cos \theta, \sin \theta), t) r d\theta.$$

The r 's cancel, so the circular mean formula becomes

$$U(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} u(x + r(\cos \theta, \sin \theta), t) d\theta.$$

This is the two-dimensional version of the spherical mean formula. In $\mathbb{R}^3$, the average is over a sphere; in $\mathbb{R}^2$, the average is over a circle.

2. Why do we introduce circular means?

The wave equation

$$u_{tt} - \Delta u = 0$$

is rotationally symmetric. This means that if we average the solution over all points at the same distance r from x , the angular dependence disappears. The original PDE in x_1, x_2, t is converted into an equation in only r and t . This is why Evans introduces spherical means.

Define

$$e(\theta) = (\cos \theta, \sin \theta).$$

Then

$$U(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} u(x + re(\theta), t) d\theta.$$

3. Compute U_{tt}

Since x and r are fixed when differentiating in t , we get

$$U_t(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} u_t(x + re(\theta), t) d\theta,$$

and hence

$$U_{tt}(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} u_{tt}(x + re(\theta), t) d\theta.$$

Because u solves the wave equation,

$$u_{tt} = \Delta u.$$

Therefore,

$$U_{tt}(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} \Delta u(x + re(\theta), t) d\theta.$$

4. Compute the averaged Laplacian

For fixed x and t , define

$$v(r, \theta) = u(x + r(\cos \theta, \sin \theta), t).$$

This is just u written in polar coordinates around the point x .

In two-dimensional polar coordinates, the Laplacian is

$$\Delta u = v_{rr} + \frac{1}{r}v_r + \frac{1}{r^2}v_{\theta\theta}.$$

Thus

$$\Delta u(x + re(\theta), t) = v_{rr}(r, \theta) + \frac{1}{r}v_r(r, \theta) + \frac{1}{r^2}v_{\theta\theta}(r, \theta).$$

Averaging over $0 \leq \theta \leq 2\pi$, we obtain

$$\frac{1}{2\pi} \int_0^{2\pi} \Delta u(x + re(\theta), t) d\theta = \frac{1}{2\pi} \int_0^{2\pi} v_{rr} d\theta + \frac{1}{r} \frac{1}{2\pi} \int_0^{2\pi} v_r d\theta + \frac{1}{r^2} \frac{1}{2\pi} \int_0^{2\pi} v_{\theta\theta} d\theta.$$

Now observe that

$$U(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} v(r, \theta) d\theta.$$

Therefore,

$$U_r(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} v_r(r, \theta) d\theta,$$

and

$$U_{rr}(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} v_{rr}(r, \theta) d\theta.$$

Also, since v is 2π -periodic in θ ,

$$\int_0^{2\pi} v_{\theta\theta}(r, \theta) d\theta = v_{\theta}(r, 2\pi) - v_{\theta}(r, 0) = 0.$$

Hence,

$$\frac{1}{2\pi} \int_0^{2\pi} \Delta u(x + re(\theta), t) d\theta = U_{rr}(x; r, t) + \frac{1}{r}U_r(x; r, t).$$

5. The Euler–Poisson–Darboux equation for $n = 2$

From the wave equation, we already had

$$U_{tt}(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} \Delta u(x + re(\theta), t) d\theta.$$

Using the computation above,

$$U_{tt}(x; r, t) = U_{rr}(x; r, t) + \frac{1}{r}U_r(x; r, t).$$

Therefore,

$$\boxed{U_{tt} - U_{rr} - \frac{1}{r}U_r = 0.}$$

This is the Euler–Poisson–Darboux equation in two dimensions.

6. Initial data

Suppose the original wave equation has initial data

$$u(x, 0) = g(x), \quad u_t(x, 0) = h(x).$$

Then the circular mean satisfies

$$U(x; r, 0) = \frac{1}{2\pi} \int_0^{2\pi} g(x + re(\theta)) d\theta =: G(x; r),$$

and

$$U_t(x; r, 0) = \frac{1}{2\pi} \int_0^{2\pi} h(x + re(\theta)) d\theta =: H(x; r).$$

Thus the original wave equation is reduced to

$$\begin{cases} U_{tt} - U_{rr} - \frac{1}{r}U_r = 0, \\ U(x; r, 0) = G(x; r), \\ U_t(x; r, 0) = H(x; r). \end{cases}$$

7. Recovering u from its circular mean

The circular mean at radius $r = 0$ is just the value at the center:

$$U(x; 0, t) = u(x, t).$$

Indeed, since

$$x + 0 \cdot e(\theta) = x,$$

we have

$$U(x; 0, t) = \frac{1}{2\pi} \int_0^{2\pi} u(x, t) d\theta = u(x, t).$$

Therefore, after studying $U(x; r, t)$, the original solution is recovered by taking

$$\boxed{u(x, t) = U(x; 0, t).}$$

8. Main conclusion

In $n = 2$, the circular mean

$$U(x; r, t) = \frac{1}{2\pi} \int_0^{2\pi} u(x + r(\cos \theta, \sin \theta), t) d\theta$$

satisfies

$$\boxed{U_{tt} - U_{rr} - \frac{1}{r}U_r = 0.}$$

The key reason is that averaging over the circle kills the angular derivative term

$$\frac{1}{r^2} v_{\theta\theta}$$

because

$$\int_0^{2\pi} v_{\theta\theta} d\theta = 0.$$

So the two-dimensional wave equation becomes a one-dimensional equation in the variables r and t .

Evans Problem 24: Equipartition of Energy

Consider the one-dimensional wave equation

$$\begin{cases} u_{tt} - u_{xx} = 0, & x \in \mathbb{R}, t > 0, \\ u(x, 0) = g(x), \\ u_t(x, 0) = h(x), & x \in \mathbb{R}. \end{cases}$$

Assume that g and h are smooth and compactly supported.

Define the kinetic energy

$$k(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_t^2(x, t) dx$$

and the potential energy

$$p(t) := \frac{1}{2} \int_{-\infty}^{\infty} u_x^2(x, t) dx.$$

We prove:

(a) $k(t) + p(t)$ is constant in t ,

and

(b) $k(t) = p(t)$ for all sufficiently large t .

1. D'Alembert formula

By D'Alembert's formula, the solution is

$$u(x, t) = \frac{1}{2} (g(x+t) + g(x-t)) + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy.$$

Differentiate with respect to t :

$$u_t(x, t) = \frac{1}{2} g'(x+t) - \frac{1}{2} g'(x-t) + \frac{1}{2} h(x+t) + \frac{1}{2} h(x-t).$$

Differentiate with respect to x :

$$u_x(x, t) = \frac{1}{2}g'(x+t) + \frac{1}{2}g'(x-t) + \frac{1}{2}h(x+t) - \frac{1}{2}h(x-t).$$

Now define

$$A(s) := g'(s) + h(s), \quad B(s) := g'(s) - h(s).$$

Then

$$u_t(x, t) + u_x(x, t) = A(x+t),$$

and

$$u_t(x, t) - u_x(x, t) = -B(x-t).$$

2. Proof that total energy is constant

We use the identity

$$a^2 + b^2 = \frac{1}{2}((a+b)^2 + (a-b)^2).$$

With $a = u_t$ and $b = u_x$, we get

$$u_t^2 + u_x^2 = \frac{1}{2}((u_t + u_x)^2 + (u_t - u_x)^2).$$

Therefore

$$u_t^2 + u_x^2 = \frac{1}{2}(A(x+t)^2 + B(x-t)^2).$$

Hence

$$k(t) + p(t) = \frac{1}{2} \int_{-\infty}^{\infty} (u_t^2 + u_x^2) dx$$

becomes

$$k(t) + p(t) = \frac{1}{4} \int_{-\infty}^{\infty} A(x+t)^2 dx + \frac{1}{4} \int_{-\infty}^{\infty} B(x-t)^2 dx.$$

Now use changes of variables. In the first integral, let

$$y = x+t.$$

Then $dy = dx$, so

$$\int_{-\infty}^{\infty} A(x+t)^2 dx = \int_{-\infty}^{\infty} A(y)^2 dy.$$

In the second integral, let

$$z = x-t.$$

Then $dz = dx$, so

$$\int_{-\infty}^{\infty} B(x-t)^2 dx = \int_{-\infty}^{\infty} B(z)^2 dz.$$

Therefore

$$k(t) + p(t) = \frac{1}{4} \int_{-\infty}^{\infty} A(y)^2 dy + \frac{1}{4} \int_{-\infty}^{\infty} B(z)^2 dz.$$

The right-hand side does not depend on t . Hence

$$\boxed{k(t) + p(t) \text{ is constant in } t.}$$

3. Proof of equipartition for large time

Now we show that

$$k(t) = p(t)$$

for all sufficiently large t .

First compute the difference:

$$k(t) - p(t) = \frac{1}{2} \int_{-\infty}^{\infty} (u_t^2 - u_x^2) dx.$$

Using

$$u_t^2 - u_x^2 = (u_t + u_x)(u_t - u_x),$$

we obtain

$$k(t) - p(t) = \frac{1}{2} \int_{-\infty}^{\infty} (u_t + u_x)(u_t - u_x) dx.$$

Substitute the formulas above:

$$u_t + u_x = A(x + t), \quad u_t - u_x = -B(x - t).$$

Thus

$$k(t) - p(t) = -\frac{1}{2} \int_{-\infty}^{\infty} A(x + t)B(x - t) dx.$$

Since g and h have compact support, both $A = g' + h$ and $B = g' - h$ also have compact support. Therefore there exists $R > 0$ such that

$$\text{supp}(A) \subset [-R, R], \quad \text{supp}(B) \subset [-R, R].$$

Now consider the product

$$A(x + t)B(x - t).$$

For this product to be nonzero, we need both

$$x + t \in [-R, R]$$

and

$$x - t \in [-R, R].$$

The first condition gives

$$x \in [-R - t, R - t],$$

and the second condition gives

$$x \in [t - R, t + R].$$

So the product can be nonzero only if the two intervals

$$[-R - t, R - t] \quad \text{and} \quad [t - R, t + R]$$

overlap.

But when $t > R$, these two intervals are disjoint, because

$$R - t < t - R.$$

Therefore, for all $t > R$,

$$A(x + t)B(x - t) = 0$$

for every $x \in \mathbb{R}$. Hence

$$k(t) - p(t) = 0.$$

Thus

$$\boxed{k(t) = p(t) \quad \text{for all sufficiently large } t.}$$

4. Interpretation

The one-dimensional wave equation separates into a right-moving wave and a left-moving wave. The functions

$$A(x + t) \quad \text{and} \quad B(x - t)$$

represent waves traveling in opposite directions.

Because g and h are compactly supported, after enough time the right-moving part and the left-moving part no longer overlap. Once they separate, the cross term

$$\int_{-\infty}^{\infty} A(x + t)B(x - t) dx$$

vanishes. This forces

$$k(t) - p(t) = 0.$$

Therefore the kinetic energy and potential energy become equal:

$$\boxed{k(t) = p(t) = \frac{1}{2}(k(t) + p(t))}$$

for all sufficiently large t .